

APPROXIMATING FIX SIZE QUANTUM CORRELATIONS IN POLYNOMIAL TIME

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Introduction

- **Motivation (Foundations):** Non-local games originate from the EPR paradox and Bell's theorem, and formalize the separation between classical and quantum correlations.
- **Quantum advantage:** Quantum strategies can outperform classical ones (e.g., CHSH [5]), corresponding to **Bell inequality violations**.
- **Computational hardness:** Even classically, approximating the value within a constant multiplicative factor is NP-hard [1, 2]. Quantumly, computing $w_{Q(T)}(G)$ is typically harder [4, 6, 9, 13], and approximation is undecidable in general [12]. For certain game families, **polynomial-time approximation** is possible [6, 13].
- **Dimension-bounded regime:** Following [11, 15–17], we restrict entanglement to fixed local dimension $|T|$ and study approximation of $w_{Q(T)}(G)$ by efficient classical algorithms.
- **Why dimension bounds matter:** Many landmark results (e.g., MIP*=RE) rely on unbounded entanglement [10, 12, 14]; with dimension bounds the complexity landscape can change drastically.

Goal & contribution: We advance the understanding of dimension-bounded quantum correlations, resolving problems posed in [11] and [3].

Main Result

For **two-player free games** with fixed question/answer alphabets and bounded local entanglement dimension, we compute ε -additive approximations to the entangled value in time $\text{poly}(1/\varepsilon)$, an **exponential speedup** in precision compared with the $\exp(1/\varepsilon)$ guarantees of prior schemes [11]. In addition, we give a rounding procedure that outputs **provably near-optimal entangled strategies**.

Extended Class of Constrained Separability Problems

$$c\text{SEP}(G) = \max_{\rho_{AB} \in \text{conv}(\mathcal{F}_A \otimes \mathcal{F}_B)} \text{Tr}[G_{AB}(\rho_{AB})],$$

$$\mathcal{F}_A := \left\{ \rho_A \succeq 0 : \text{Tr} \rho_A = 1, \Theta_{A_L \rightarrow C_{A_L}}[\rho_A] = W_{C_{A_L}} \otimes \rho_{A_R}, \Omega_{A \rightarrow A}[\rho_A] = \rho_A \right\},$$

$$\mathcal{F}_B := \left\{ \rho_B \succeq 0 : \text{Tr} \rho_B = 1, Y_{B_L \rightarrow C_{B_L}}[\rho_B] = K_{C_{B_L}} \otimes \rho_{B_R}, \Xi_{B \rightarrow B}[\rho_B] = \rho_B \right\},$$

where $\Theta_{A_L \rightarrow C_{A_L}}, Y_{B_L \rightarrow C_{B_L}}, \Omega_{A \rightarrow A}, \Xi_{B \rightarrow B}$ are linear maps and $G_{AB}, W_{C_{A_L}}, K_{C_{B_L}}$ are constant operators.

We generalize the approximation schemes of [3] to handle richer constraint families, substantially **expanding the set of applications** that can be tackled within this framework.

Information-theoretic Bose-symmetric de Finetti

- **Stronger notion of Symmetry:** States w/ support & range on symmetric subspace

$$[\mathbb{1}_A \otimes U_{B_1^n}(\pi)] \rho_{AB_1^n} = \rho_{AB_1^n}, \forall \pi \in S_n$$

- **Constraint maps:** Separable states not simply representable as convex mixtures of pure product states
- **Standard de Finetti arguments fail:** They do not enforce linear constraints on each mixture component. We extend the **information-theoretic proof methods** of [3, 4] to the Bose-symmetric setting

Lemma (Constrained Bose-symmetric de Finetti theorem). *Let $\rho_{A(B\bar{B})_1^n}$ be Bose-symmetric with respect to A . Assuming linear mappings*

$$\Theta_{A_L \rightarrow C_{A_L}}(\rho_{A(B\bar{B})_1^n}) = W_{C_{A_L}} \otimes \rho_{A_R(B\bar{B})_1^n}, \quad \Omega_{A \rightarrow A}(\rho_{A(B\bar{B})_1^n}) = \rho_{A(B\bar{B})_1^n},$$

$$Y_{(B_L)_n \rightarrow C_{B_L}} \circ \text{Tr}_{\bar{B}_n}[\rho_{(B\bar{B})_1^n}] = K_{C_{B_L}} \otimes \rho_{(B\bar{B})_1^{n-1}(B_R)_n}, \quad \Xi_{B_n \rightarrow B_n} \circ \text{Tr}_{\bar{B}_n}[\rho_{(B\bar{B})_1^n}] = \rho_{(B\bar{B})_1^{n-1}B_n},$$

then there exists a probability distribution $\{p(x)\}_{x \in \mathcal{X}}$ and states $\{\sigma_A^x\}_{x \in \mathcal{X}}, \{\sigma_{B\bar{B}}^x\}_{x \in \mathcal{X}}$ such that

$$\left\| \rho_{A(B\bar{B})} - \sum_{x \in \mathcal{X}} p(x) \sigma_A^x \otimes \sigma_{(B\bar{B})}^x \right\|_1 \leq \mathcal{O}\left(\frac{|\mathcal{H}_B \mathcal{H}_{\bar{B}}|}{\sqrt{n}} \cdot \sqrt{\ln |\mathcal{H}_A|}\right)$$

and for each $x \in \mathcal{X}$

$$\Theta_{A_L \rightarrow C_{A_L}}(\sigma_A^x) = W_{C_{A_L}} \otimes \sigma_{A_R}^x, \quad \Omega_{A \rightarrow A}(\sigma_A^x) = \sigma_A^x,$$

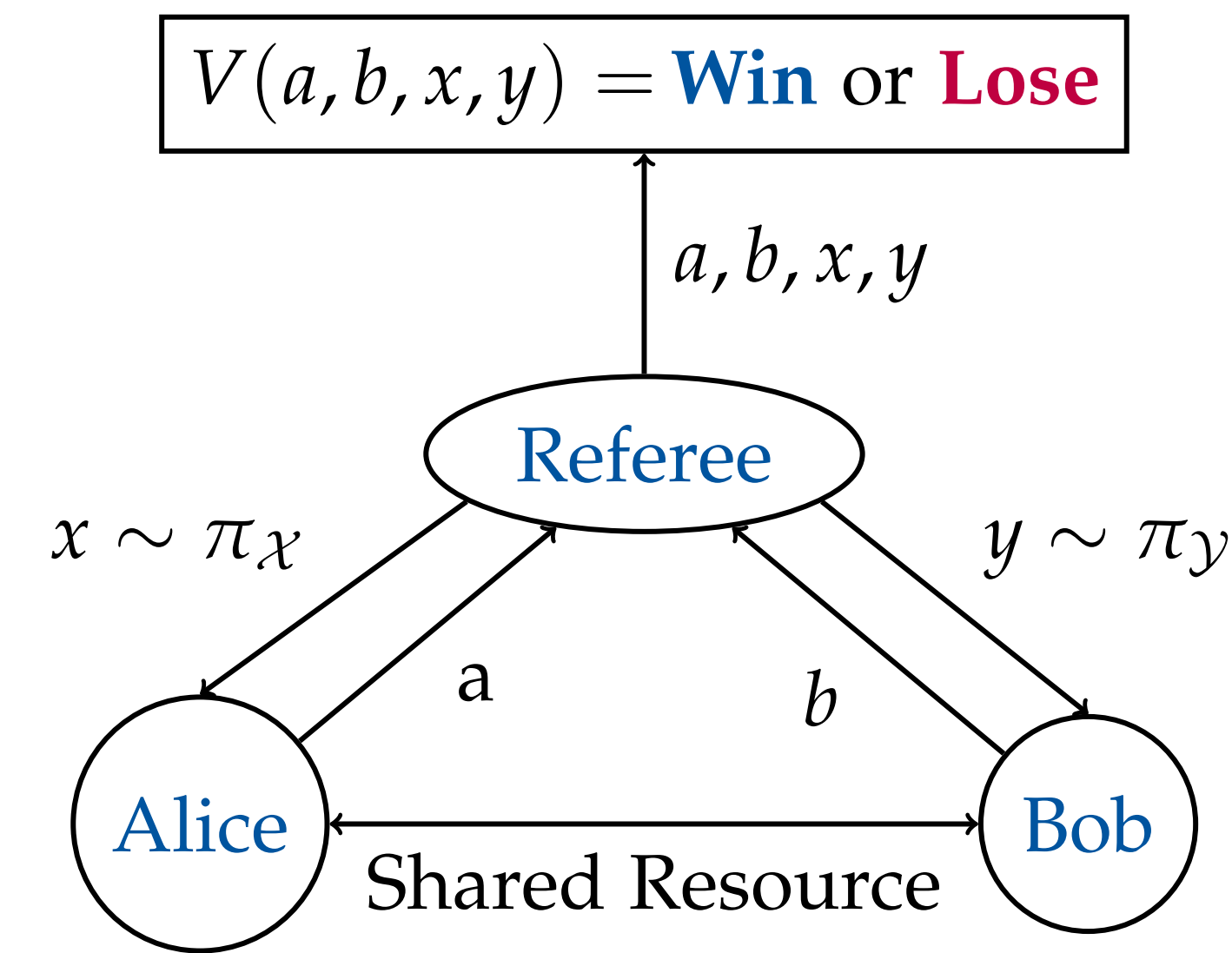
$$Y_{B_L \rightarrow C_{B_L}} \circ \text{Tr}_{\bar{B}}[\sigma_{(B\bar{B})}^x] = K_{C_{B_L}} \otimes \sigma_{B_R}^x, \quad \Xi_{B \rightarrow B_n} \circ \text{Tr}_{\bar{B}}[\sigma_{(B\bar{B})}^x] = \sigma_B^x.$$

Symmetric-subspace techniques [7, 8] do extend to constrained separability.

This work: [arXiv:2507.12302](https://arxiv.org/abs/2507.12302)

and GPTs [arXiv:2601.15184](https://arxiv.org/abs/2601.15184):

Non-local Games



Game model: We consider *two-player free non-local games* $G = (\mathcal{A}, \mathcal{B}, \mathcal{X}, \mathcal{Y}, V, \pi)$, where questions $x \sim \pi_{\mathcal{X}}, y \sim \pi_{\mathcal{Y}}$ are sampled *independently* and the referee accepts answers a, b with probability $V(a, b, x, y)$. **No communication:** Players may pre-agree on a strategy, but *cannot communicate* after receiving x, y .

Strategies:

– *Classical:* Deterministic/stochastic response functions

– *Quantum:* A shared entangled state of bounded dimension $|T|^2$ plus local complete POVMs conditioned on the questions

Objective: Maximize the winning probability:

$$w_{Q(T)}(G) = \max_{\text{strategies}} \sum_{a,b,x,y} \pi_{\mathcal{X}}(x) \pi_{\mathcal{Y}}(y) V(a, b, x, y) p(a, b | x, y)$$

Polynomial Time Algorithm from Bose-symmetry

- **Local purification scheme:** Careful embedding of constraint maps into higher-dimensional symmetric subspace
- **Convergent SDP hierarchy** $SDP_n^{\text{Bose}}(G)$ via Bose-symmetric extensions approximating cSEP
- **End-to-end efficiency:** Polynomial-time algorithm requires polynomial SDP size and polynomial-time constructibility of SDP
- **Symmetry-adapted construction:** Schur–Weyl duality to build the SDP directly in symmetry-adapted basis.

Theorem. *For a two-player free non-local game with $|A|$ answers, $|Q|$ questions and quantum assistance of size $|T|$, there exists a $\text{poly}(1/\varepsilon)$ costly transformation to an SDP with*

$$(\varepsilon^{-4} \cdot \text{poly}(|A|, |Q|, |T|))^{(|A||Q||T|)^2} \quad (1)$$

many variables, which approximates the value of the game up to an additive ε -error.

We present an **easy-to-implement** cSEP approximation algorithm with an **exponential speedup** in precision – i.e., polynomial runtime in $(1/\varepsilon)$ for additive error ε – over all previously known approaches [3, 11].

Convergent Inner Sequence

Measurement-based rounding procedure for cSEP with de Finetti-based **convergence guarantees**

Theorem. *There exists a rounding procedure that converts an ε -good approximation obtained from $SDP_n^{\text{Bose}}(G)$ of $w_{Q(T)}(G)$ from above into an ε -good approximation from below corresponding to a feasible strategy of Alice's and Bob's sets of complete POVMs and a shared quantum state of fixed dimension.*

We complete the approximation landscape by giving a **convergent inner-approximation sequence** for cSEP that outputs constrained separable states together with **provable approximation guarantees**

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